SOFTWARE REQUIREMENTS SPECIFICATION

for

ClassLink

A BLE Based Classroom Attendance System

Version 1.0

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Document status: the backend is implemented and tested. The Android client integration is planned. Section 3.2 records the status of every requirement.

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1 Introduction

1.1 Purpose

Taking attendance by roll call eats lecture time, and a signed sheet or a shouted "present" is easy to fake and hard to check later. ClassLink is the fix. It is a classroom attendance system where the teacher's phone broadcasts a short lived Bluetooth Low Energy (BLE) token, each student's phone hears it and sends a signed claim, and a server decides who counts as present. The main idea is that the phone never decides for itself, so the record can be trusted.

This document is the Software Requirements Specification (SRS) of ClassLink Version 1.0. It follows the approved project proposal (UCS503P). Requirements carry an ID such as FR (functional) and a priority from the MoSCoW scheme: Must (needed for the pilot), Should (next iteration), Could (optional). A Status column says what is built today.

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, testers, pilot teachers, and the course instructor. It gives the internal, external, functional, and nonfunctional details of ClassLink. Every requirement is written so a test can check it.

Key terms used in this document:

- **BLE (Bluetooth Low Energy)**: the short range radio the phones use. It needs no pairing.
- **Token**: a 17 byte code the teacher phone broadcasts. It changes every 15 seconds and names no person.
- **Challenge and nonce**: a one time value the server gives a student before a check in, so a claim cannot be replayed.
- **Claim**: the signed message a student phone sends. It holds the nonce, token, signal strength, sample count, and time.
- **RSSI**: received signal strength in dBm. A larger number (closer to zero) means a closer phone.
- **Session**: one class period in which attendance is open.
- **Device key**: the signing key kept on a student's phone. The server stores only its public half.
- **Audit log**: the server's permanent record of security relevant actions.
- **POC**: the earlier offline proof of concept app.

1.3 Project Scope

ClassLink gives Teachers and Students one place to run attendance, and gives the Administrator control over who may teach. It records presence only. It does not grade, schedule, or track location.

Objectives.

1. Speed: a student checks in within 10 seconds of starting the scan, so roll call stops costing lecture time.
2. Trust: the server verifies every claim, so a phone cannot mark itself present.
3. Auditability: every claim, rejection, and correction leaves a record.

4. Teacher control: a live roster, manual corrections with a reason, and CSV export.
5. Privacy: the system keeps only name, roll number, email, a device public key, and attendance. It never collects location.
6. Engineering quality: the backend passes Ruff and strict MyPy, and every requirement marked Done has an automated test run in CI.

In scope. Teacher and student roles on Android. Classrooms with join codes. One registered phone per student. Attendance sessions with a signal strength rule. A live roster, manual corrections, and CSV export. An audit log. Testing on real phones in a real room.

Optional extensions (outside this course submission). An iOS app, links to learning management systems, and a signal threshold that adapts to each room.

Out of scope. Proving that the phone's owner is the person holding it. Student to student relaying. Grading and timetables.

Status terms. *Done* means built and covered by an automated test. *POC* means present only in the offline proof of concept. *Planned* means specified but not built. Today the server side is built and tested. The Android app does not call it yet, so anything that needs the app is Planned.

Stakeholders.

Table 1.1: Stakeholders of ClassLink.

Stakeholder	Role	Main interest
Students	Join classrooms, register a phone, check in	A quick check in that counts
Teachers	Create classrooms, run sessions, correct records	Accurate attendance without roll call
Administrator	Deploys the server, edits the teacher allowlist	A secure, running system
Development team	Design, build, test, deploy	Clear, testable requirements
Course instructor	Reviews and evaluates the project	Consistency with the approved proposal
Supabase	Hosts the database and sign in	Fair use of its free tier

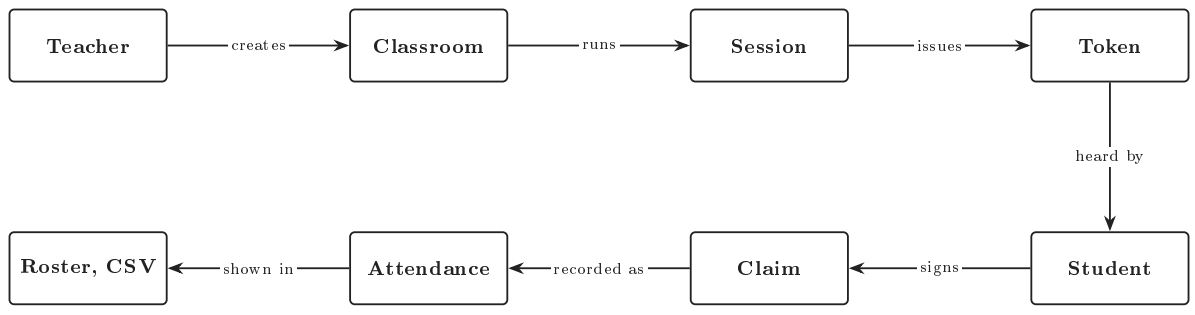


Figure 1.1: Entire work-flow.

Figure 1.1 is the overview of the project. A Teacher creates a Classroom. The Classroom runs a Session. The Session issues rotating Tokens. A Student hears a Token and signs a Claim. The server checks the Claim into an Attendance record, and the Teacher sees it in the Roster and CSV. Every entity feeds the next one, so the whole system works as one chain.

2 Overall Description

2.1 Product Perspective

ClassLink grew from an offline proof of concept, the Flutter app in `code/`, that showed a teacher phone can announce a session over BLE and a student phone can check in by proximity. Version 1.0 adds a server so the result no longer depends on trusting the student's phone. It follows a layered client and server design (Figure 2.1).

- **Mobile app:** Flutter, Android first, with a teacher mode and a student mode.
- **Backend API:** FastAPI (Python) for sign in checks, validation, sessions, and claims.
- **Business logic:** rotating tokens, claim checks, corrections, and export.
- **Database:** Supabase Postgres. Supabase Auth signs users in.

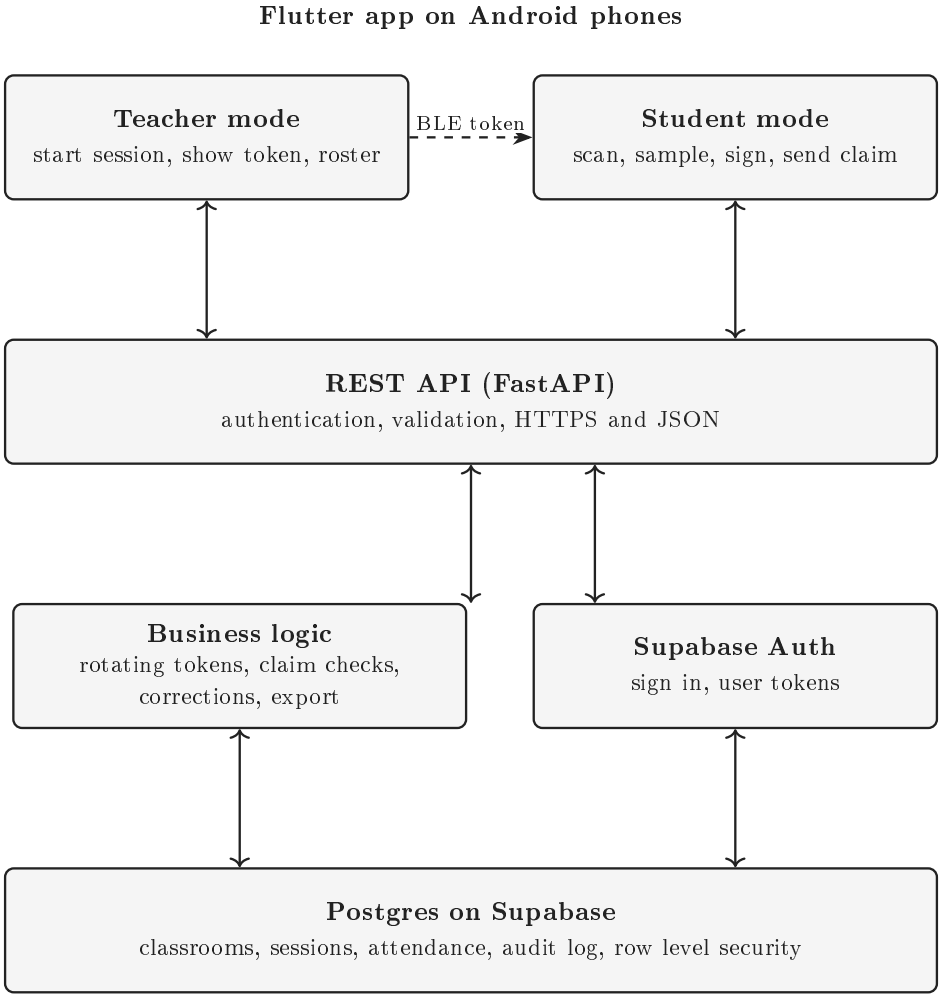


Figure 2.1: High-level system architecture.

One check in works like this. The teacher app fetches a fresh token from the server every 15 seconds and broadcasts it. A student phone hears the token and asks the server for a one time challenge. The phone measures signal strength for a few seconds, signs the result with its own key, and sends it. The server verifies the claim and records attendance.

2.2 User Classes and Characteristics

ClassLink has four user classes (Figure 2.2).

- **Teacher.** Runs sessions for classrooms they own. The role comes only from an email allowlist kept by the administrator.
- **Student.** Joins classrooms with a join code and checks in with one registered phone.
- **Administrator.** Deploys the server and edits the teacher allowlist. This is not an in app role.
- **Evaluator.** Reads the exported CSV files. No app access needed.

Supabase Auth is the one external system. It signs users in and issues their tokens.

Students and teachers need only basic phone skills. The administrator needs medium technical skill.

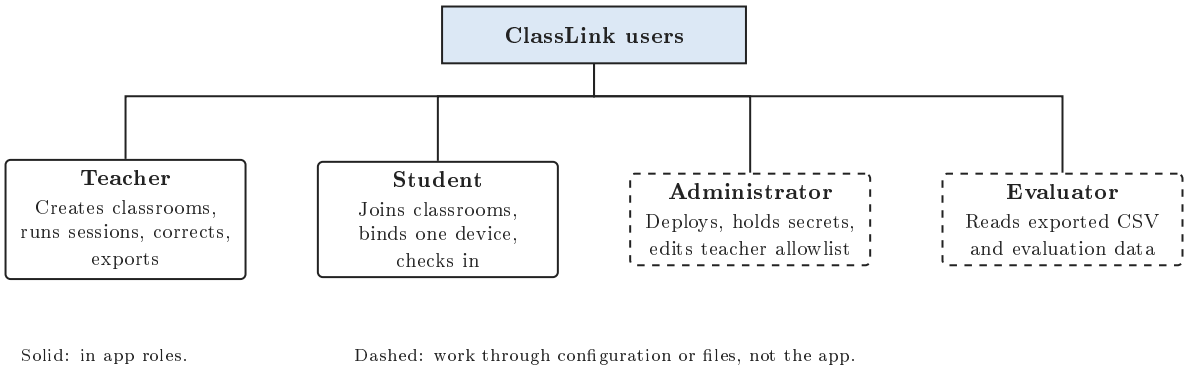


Figure 2.2: Type of users.

2.3 Product Functions

ClassLink stores classrooms, sessions, and attendance, and keeps an audit trail. At a high level it lets users sign in and get a role, create or join a classroom, register a device key, start and end sessions with rotating tokens, check in with a signed claim, watch a live roster, correct mistakes with a reason, and export a CSV. Users must sign in before using any of these functions.

The principal data entities are:

- **profiles**: user id, role (teacher or student), full name, roll number, email.
- **classrooms**: classroom id, teacher id, name, unique join code.
- **enrollments**: classroom id and student id, with the join time.
- **devices**: student id, public key, label, revoked time. A student has one active device.
- **sessions**: classroom id, alias, signal threshold, sample window, minimum samples, start and end time. A classroom has at most one active session.
- **challenges**: session id, student id, token, issue and expiry time, used time. Each is single use.
- **attendance**: session id, student id, status, source (BLE claim or manual correction), median RSSI, sample count, observation time, and correction details. One record per student per session.
- **audit_events**: actor, action, subject, details, time.

Foreign keys keep the data consistent. The level 0 diagram (Figure 2.3) treats ClassLink as one box and shows what crosses its edge. Arrows pointing right mean data going in. Arrows pointing left mean data coming out.

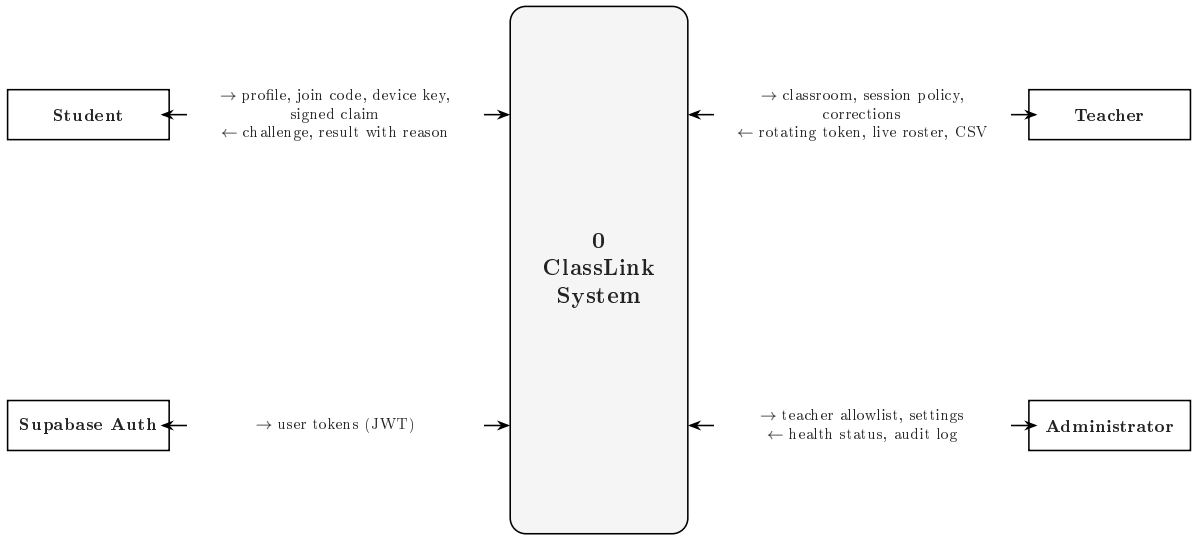


Figure 2.3: Data flow diagram, level 0.

Level 1 (Figure 2.4) opens the box into six processes. Each process keeps its data in a store, drawn under it.

2.4 Operating Environment

Students and teachers use Android 8.0 or later with Bluetooth LE. The teacher phone must support BLE advertising, so we test on real phones. The server runs in a Linux container behind HTTPS. The database is Supabase Postgres. Phones need internet during check in, but the Bluetooth exchange itself does not.

External interfaces.

- **User interface:** a Flutter app with a teacher mode (classrooms, session control, live roster, corrections, export) and a student mode (join, register phone, check in, result). Every rejection shows a plain explanation.
- **Hardware:** Bluetooth LE radio. The teacher phone needs BLE advertising. No other hardware.
- **Software:** FastAPI backend, Supabase Postgres, Supabase Auth, GitHub Actions for CI.
- **Communications:** HTTPS with JSON over REST. A Supabase JWT in the Authorization header. A 17 byte token over BLE manufacturer data.

Design and implementation constraints.

- Stack: Flutter (Android first), FastAPI on Python 3.11 or later, psycopg 3, Supabase Postgres and Auth.
- A phone never creates an attendance record on its own. Only the server decides.
- Secrets stay on the server. Only `.env.example` files enter the repository.
- The BLE token is 17 bytes and travels as manufacturer data, because a service identifier plus the token does not fit in a basic advertisement. We have not yet confirmed this layout on phones.

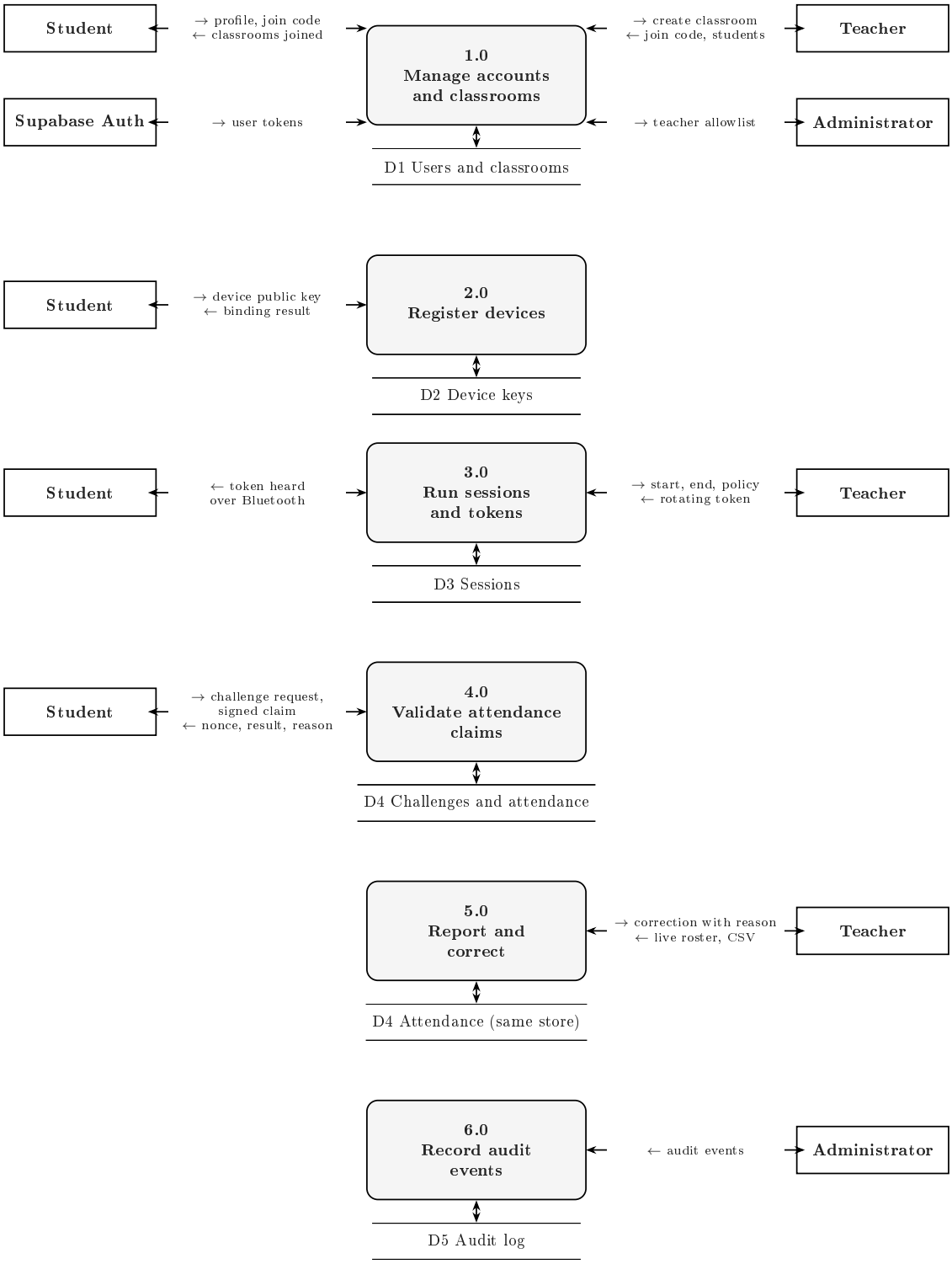


Figure 2.4: Data flow diagram, level 1.

- Apps write only through the API. The database gives apps read only access to their own rows.
- Four person team, one semester, with the work split across backend, app, documentation, and testing.

Assumptions and dependencies.

- Students use the phone they registered. A student who changes phones registers again.
- Phone and server clocks stay within about a minute.
- Supabase free tier limits are enough for the pilot.
- Signal strength is noisy, so the server uses the median of several samples against a per session threshold. The right threshold is tuned on real phones.

2.5 Design

A check in follows the flow in Figure 2.5. It crosses three places: the teacher app, the server, and the student app.

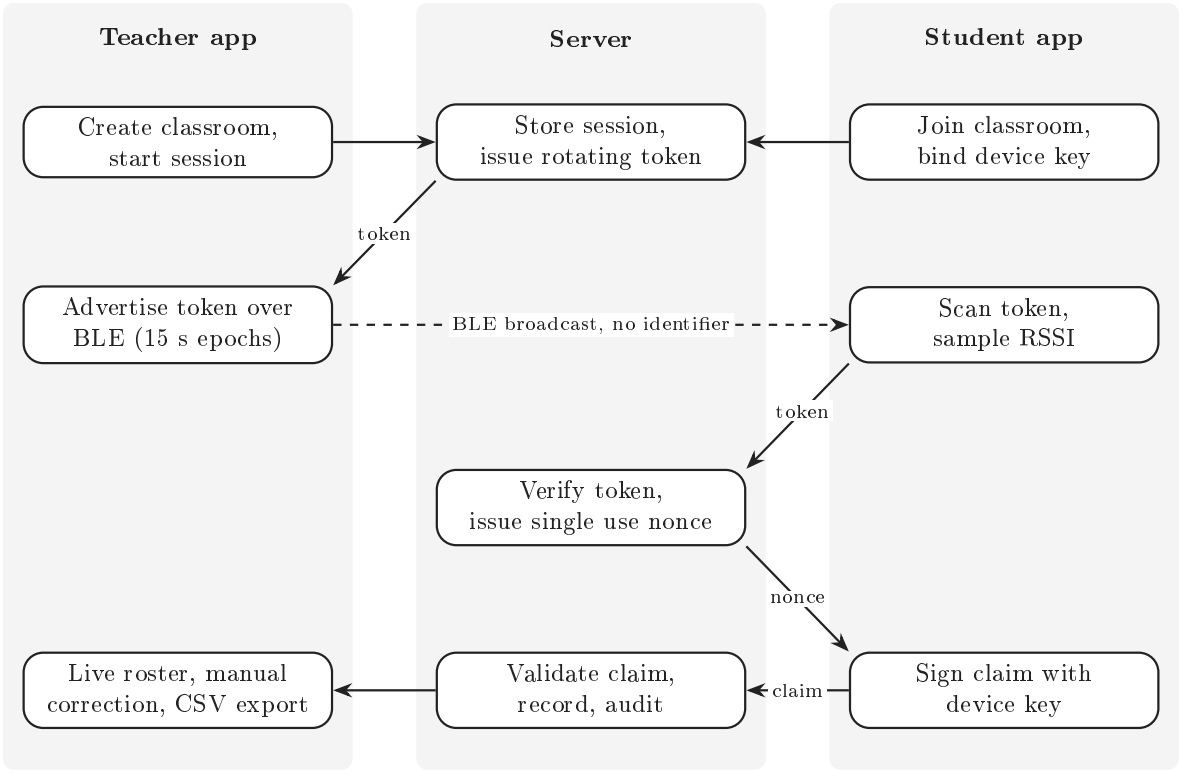


Figure 2.5: Check in workflow.

Student activities have three steps: join a classroom, register the phone once, and check in. Everything ends in the student's profile (Figure 2.6).

Teacher activities (Figure 2.7) cover setting up a classroom, running a session, and keeping the records straight.

The administrator has one small job. They keep the server running and decide who is a teacher (Figure 2.8).

Figure 2.9 shows every use case and who starts it.

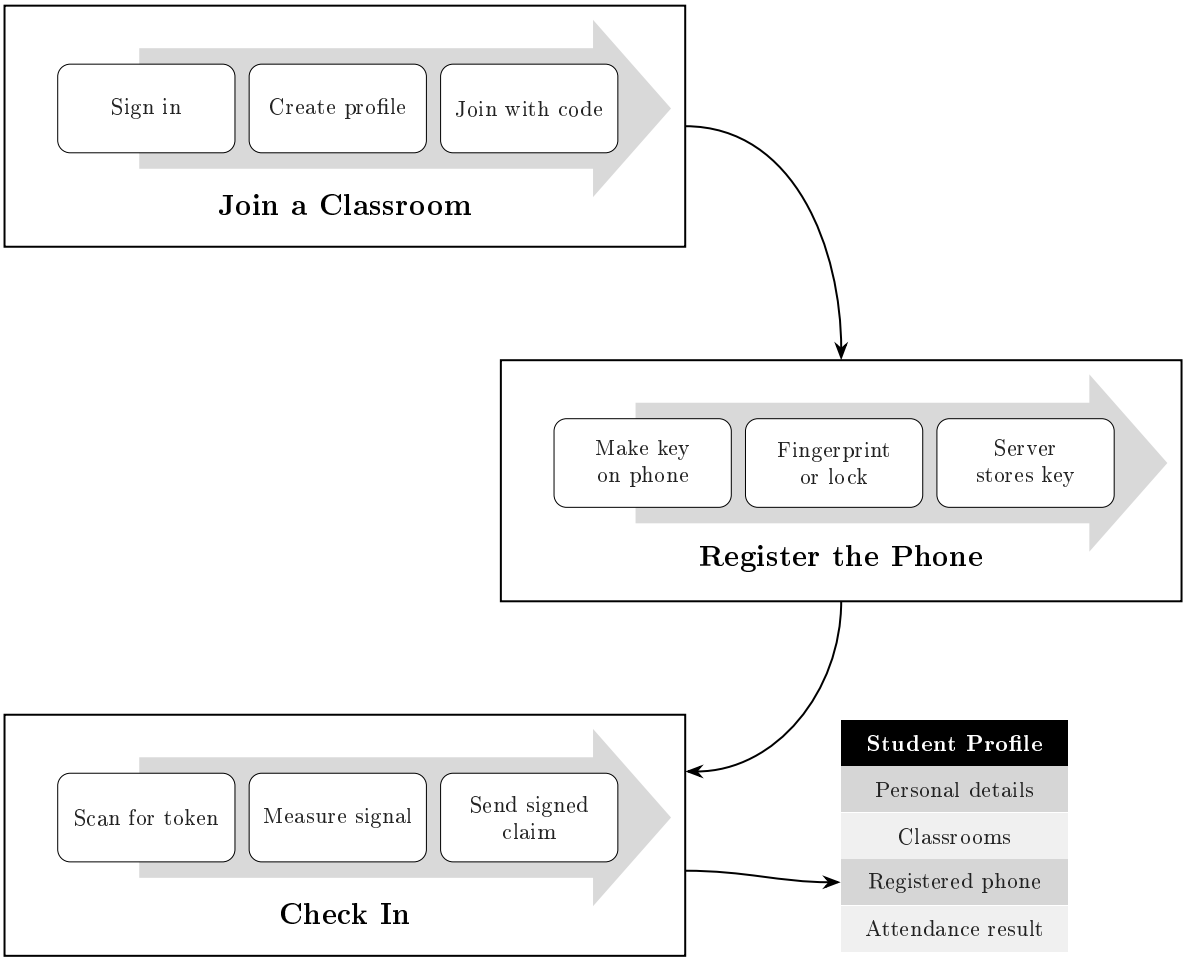


Figure 2.6: Student activities.

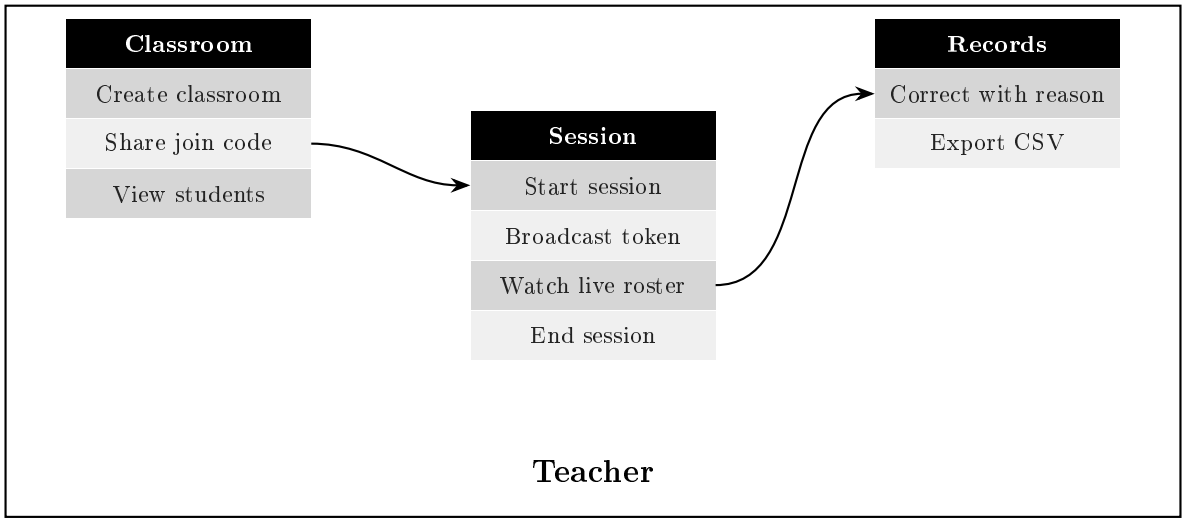


Figure 2.7: Teacher activities.



Figure 2.8: Administrator activities.

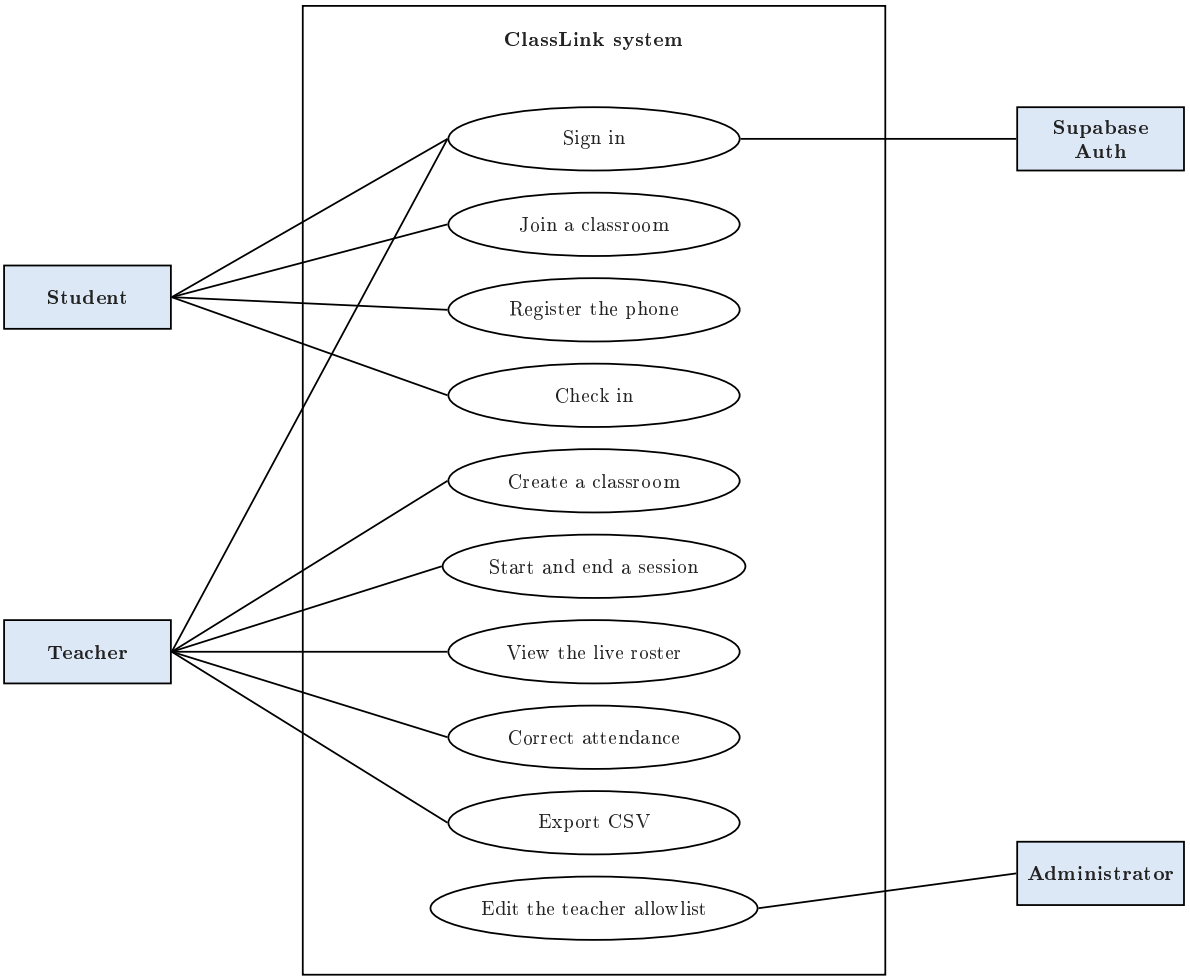


Figure 2.9: Use case diagram (Student, Teacher and Administrator activities).

UC-1 Check in to a session. Actor: student, who must be signed in, enrolled, and have a registered phone. The teacher has an active session. The student taps Check in. The app scans for the token for the sample window and requests a challenge. The server checks the token, session, enrollment, and device, then issues a single use nonce. The app signs the claim and sends it. The server checks the nonce, signature, time, sample count, and median signal, then records attendance and shows Present. If the signal is weak or samples are too few, the student sees a plain message and may move closer and retry. Any other failure shows its reason code and nothing is recorded.

UC-2 Correct attendance. Actor: teacher, who owns the classroom. The teacher opens the roster, picks a student, and marks them present or absent with a reason of at least five characters. The server stores the change with the teacher's name and writes an audit event. A reason that is too short is refused and nothing changes.

The design rules behind these flows:

- **Teacher broadcasts, student listens.** There is no relay between students.
- **The server decides.** Every claim is checked on the server.
- **Real data only.** Evaluation figures come only from recorded real runs.

3 System Features

ClassLink is a classroom attendance system. The main part of this product is to accept a signed BLE claim and decide on the server whether it counts.

3.1 Description and Priority

ClassLink has main features and some sub features. In priority order, from top to bottom:

1. **Attendance claims** (Must, pilot): challenge, signed claim, server checks. This is the goal feature of the software.
2. **Sessions and tokens** (Must, pilot): start and end sessions, rotating BLE token.
3. **Device binding** (Must, pilot): one registered key per student.
4. **Accounts and classrooms** (Must, pilot): sign in, roles, join codes.
5. **Roster and corrections** (Must, pilot): live list, fixes with a reason.
6. **Export and audit** (Should): CSV download and audit log.
7. **Offline mode** (Could): the original no server demo, labelled as not authoritative.

The first meaningful milestone is a working pilot where a teacher starts a session, a student checks in from a real phone, and the roster shows the result. The server side of this is done. The app side is planned.

3.2 Functional Requirements

The server is a FastAPI service built and tested against a real Postgres. The app still needs its API client, sign in, key storage, token broadcast, and roster screens. Priority is Must, Should, or Could. The Status column marks what is built.

Accounts and classrooms

ID	Requirement	Priority	Status
FR-ACC-01	The API shall accept only requests with a valid, unexpired Supabase token and answer all others with HTTP 401.	Must	Done
FR-ACC-02	A user shall have one profile with the role teacher or student. Only emails on the server allowlist become teachers. A student profile shall require a roll number.	Must	Done
FR-ACC-03	Students shall not reach teacher operations, and teachers shall not reach student operations (HTTP 403).	Must	Done
FR-CLS-01	A teacher shall create a classroom and receive a unique six character join code.	Must	Done
FR-CLS-02	A student shall join a classroom with its code. An unknown code shall return HTTP 404.	Must	Done

Device binding

ID	Requirement	Priority	Status
FR-DEV-01	A student shall register one device key (ECDSA P-256 public key). The server shall reject any other key type.	Must	Done
FR-DEV-02	Registering a new key shall revoke the old one and write an audit event.	Must	Done
FR-DEV-03	The private key shall never leave the phone, and the app shall sign only after fingerprint or screen lock authentication.	Must	Planned

Sessions and tokens

ID	Requirement	Priority	Status
FR-SES-01	A teacher shall start a session for their own classroom with a signal threshold (default −75 dBm), a sample window (default 4 s), and a minimum sample count (default 3). A classroom shall have at most one active session.	Must	Done
FR-SES-02	A teacher shall end a session. After that the server shall reject its tokens, challenges, and claims.	Must	Done
FR-SES-03	The server shall issue a 17 byte rotating token that changes every 15 seconds and carries no student or teacher identity.	Must	Done
FR-SES-04	Only the classroom's teacher shall see or change its sessions. Other teachers shall get HTTP 404.	Must	Done
FR-SES-05	The teacher app shall fetch a fresh token each period and broadcast it.	Must	Planned

ID	Requirement	Priority	Status

Attendance claims

ID	Requirement	Priority	Status
FR-ATT-01	The student app shall scan for the token and collect signal strength samples.	Must	Planned
FR-ATT-02	On a challenge request the server shall check the token, the active session, enrollment, and the student's registered device, then issue a single use nonce valid for 120 seconds.	Must	Done
FR-ATT-03	A claim shall hold the nonce, token, median signal strength, sample count, and observation time (whole Unix seconds), signed with the device key over a fixed text format.	Must	Done
FR-ATT-04	The server shall accept a claim only if the nonce is the caller's, unused, and unexpired, the token matches, the signature verifies, the observation time is within 60 seconds of server time, the sample count meets the minimum, and the median signal meets the threshold.	Must	Done
FR-ATT-05	A student shall have at most one attendance record per session. Repeating an accepted claim shall return the same record.	Must	Done
FR-ATT-06	A weak signal or too few samples shall leave the nonce usable so the student can move closer and retry.	Should	Done
FR-ATT-07	Every rejection shall carry a reason code (listed in the supplement) with HTTP 409.	Must	Done
FR-ATT-08	The student app shall show the result and explain each reason in plain words.	Must	Planned

Roster, corrections, export, audit

ID	Requirement	Priority	Status
FR-ROS-01	A teacher shall see every enrolled student of a session with status, source, signal strength, and time.	Must	Done
FR-ROS-02	The teacher app shall refresh the roster at least every 5 seconds during a session.	Should	Planned
FR-COR-01	A teacher shall mark a student present or absent with a reason of at least five characters. The server shall audit each correction.	Must	Done
FR-EXP-01	A teacher shall download a session as CSV with every enrolled student. The export shall neutralise cells that start with `=`, `+`, `-`, or `@`.	Must	Done
FR-AUD-01	The server shall audit classroom creation, device binding, session start and end, accepted and rejected claims, corrections, and exports. Apps shall have no access to the audit log.	Must	Done
FR-POC-01	The original offline mode shall remain and be labelled as not authoritative.	Could	POC

4 Other Nonfunctional Requirements

4.1 Performance Requirements

These are targets. We report only values measured in the pilot.

- Discovery: a student phone shall pick up the token within 5 seconds in the same room.
- Check in: from scan start to server result within 10 seconds.
- API: p95 under 500 ms for challenge and claim calls with 50 students at once.
- Capacity: one classroom of 300 students with up to 150 check ins at once.

4.2 Security Requirements

- All traffic shall use HTTPS in production.
- Secrets, such as the token key and database credentials, shall exist only on the server. CI shall fail if a `.env` file or keystore is committed.
- The database shall have row level security on every table. Apps read only their own rows and write nothing.
- A signed claim shall cover the session, nonce, token, signal strength, sample count, and time, so changing any value breaks the signature.
- The server shall check token tags in constant time and verify every request's token signature, expiry, and audience.
- The system shall store only name, roll number, email, a device public key, and attendance. It shall not collect location.

ClassLink resists casual proxy attendance. It does not stop a friend who holds the student's unlocked phone, or someone who relays a live token to a faraway phone within its few seconds of life. Short token periods, the signal rule, and the audit log reduce these risks but do not remove them.

4.3 Software Quality Attributes

- **Reliability:** attendance writes are atomic and safe to retry. Two racing claims for one student produce one record.
- **Availability:** `/healthz` and `/readyz` report server and database health.
- **Maintainability:** the backend passes Ruff and strict MyPy in CI, and every Done requirement has an automated test.
- **Portability:** the server runs from one container image configured by environment variables.
- **Usability:** a student checks in with one tap, and every rejection has a plain explanation.

Metric	Target

Acceptance criteria.

Table 4.1: Acceptance criteria.

Metric	Target
Discovery	A phone picks up the token within 5 seconds in the same room
Check in time	Scan start to server result within 10 seconds
API speed	p95 under 500 ms for challenge and claim calls with 50 students at once
Functional completeness	All Must requirements demonstrated end to end on real phones
Test coverage	Every requirement marked Done has an automated test that passes in CI
Secret safety	CI fails if a `.env` file or keystore is committed
Honest results	Pilot figures come only from recorded real runs

4.4 Business Rules

- A student has at most one attendance record per session.
- Only a classroom's teacher can see or change its attendance.
- A manual correction always has a reason and a named teacher.
- An ended session never restarts. A new session gets new tokens.
- A student has one active phone. Replacing it revokes the old key.
- Only the administrator decides who becomes a teacher.

5 Other Requirements

ClassLink needs maintenance after the course, because phones, Android versions, and BLE behaviour keep changing. Other requirements are:

- **Documentation:** the repository holds this SRS in Markdown and LaTeX, the project proposal, UML diagrams, the deployment runbook, and the API reference served at `/docs`.
- **Deployment:** the server builds into a container with a non root user and refuses to start in production without its database URL and secrets. The administrator applies the migration file to Supabase before the first start. GitHub Actions runs the app and server checks and rejects committed secrets. The team publishes a signed Android APK for the pilot. See `docs/deployment.md`.
- **Privacy:** pilot participants learn what the app stores before they join a classroom. Deleting a user deletes their profile, enrollments, devices, and attendance. English only in the first version.
- **Phasing:** Phase 1 (done): server, database schema, tests, CI, and documents. Phase 2 (planned): Android client with sign in, key storage, token broadcast, scan and claim, and roster screens. Phase 3 (planned): pilot on real phones in a real room, with measured results. Extensions follow only after the pilot.
- **Open issues to confirm:** the BLE manufacturer data layout on real phones; the signal threshold that works in a real room; whether to move from a shared secret JWT to asymmetric keys; the Android application id, which still carries the old proof of concept name; and the consent wording for pilot students.

Risks and mitigations.

Risk	Mitigation
Some phones cannot advertise BLE	Test on real phones early and list supported models
Signal strength varies by room and phone	Median of several samples, a per session threshold, and a manual correction path
A friend relays a live token	Short token life, one time nonce, the signal rule, and the audit log
Server or database downtime	Health endpoints, a single container image, and managed Postgres
Leaked secrets	Secrets only on the server, and a CI check that rejects committed ones
Scope creep	Keep extensions out of the pilot, and track status in Section 3.2

References. (1) ClassLink Project Proposal, UCS503P, Thapar Institute of Engineering and Technology. (2) IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications. (3) Supabase and FastAPI documentation.